Now suppose we compute the average velocities over shorter and shorter time intervals $[t_0, t_0 + \Delta t]$. In other words, we let Δt approach 0. Then we define the velocity (or instantaneous velocity) $v(t_0)$ at time $t = t_0$ as the limit of these average velocities.

$$v(t_0) = \lim_{\Delta t \to 0} \frac{f(t_0 + \Delta t) - f(t_0)}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}.$$

This means that the velocity at time $t = t_0$ is equal to the slope of the tangent line at P.

Illustration 10.2

The distance s travelled by a body falling freely in a vacuum and the time t of descent are variables. They depend on each other. This dependence is expressed by the law of the free fall :

$$s = \frac{1}{2} g t^2 \text{ (absence of initial velocity), } g \text{ is the gravitational constant.}$$

Step (i) $f(t_0 + \Delta t) = \frac{1}{2} g (t_0 + \Delta t)^2 = \frac{1}{2} g \left(t_0^2 + 2 t_0 \Delta t + (\Delta t)^2 \right)$

Step (ii) $\Delta s = f(t_0 + \Delta t) - f(t_0)$

$$= \frac{1}{2} g \left[(t_0^2 + 2 t_0 \Delta t + (\Delta t)^2 \right] - \frac{1}{2} g t_0^2$$

$$= g \, \Delta t \left[t_0 + \frac{1}{2} \Delta t \right]$$



Step (iii) $\frac{\Delta s}{\Delta t} = \frac{g \, \Delta t \left[t_0 + \frac{1}{2} \Delta t \right]}{\Delta t} = g \left[t_0 + \frac{1}{2} \Delta t \right]$

Step (iv) $v(t_0) = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = g t_0$.

It is clear from this that the velocity is completely defined by the instant t_0. It is proportional to the time of motion (of the fall).

10.2.3 The derivative of a Function

We have now arrived at a crucial point in the study of calculus. The limit used to define the slope of a tangent line or the instantaneous velocity of a freely falling body is also used to define one of the two fundamental operations of calculus – **differentiation**.

Definition 10.2

Let f be defined on an open interval $I \subseteq$ containing the point x_0, and suppose that $\lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ exists. Then f is said to be **differentiable** at x_0 and the **derivative** of f at x_0, denoted by $f'(x_0)$, is given by

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

For all x for which this limit exists,

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \text{ is a function of } x.$$

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